Designing Visual Tools for Writing Process Analysis

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Abstract

Understanding how texts are produced is crucial not only for the development of theoretical models and writing strategies, but also for practical applications. However, the writing process itself—including intermediate versions, copy-paste actions, input from co-authors or LLMs—remains invisible in the final text. This study addresses this gap by visualizing fine-grained keystroke logging data to capture both the product (final text) and the process (writer's actions) at sentence and text level. We design and implement custom JavaScript visualizations of linguistically processed keystroke logging data. Our pilot study examines data from nine students writing under identical conditions; we analyze temporal, spatial, and structural aspects of writing. The results reveal diverse, non-linear writing strategies and suggest that individualized process visualizations can inform both document engineering and writing analytics. The novel visualization types we present demonstrate how process and product can be meaningfully integrated.

CCS Concepts

• **Human-centered computing → Visualization techniques**; *Visual analytics*; **Visualization design and evaluation methods**; • **Applied computing** → *Text editing*.

Keywords

Writing Process, Keystroke Logging, Visualization, Writing Research

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1 Introduction

Producing texts leaves traces. The final text, however, usually renders invisible everything that happened during the writing process. Replaced words, rearranged sentences, typos—they are all gone. Writing process research using keystroke logging makes these traces visible. All actions are recorded and can be evaluated and analyzed. Writing process data can be evaluated in different ways and presented. The history of the creation of the entire text—the text history—can be told or the history of the creation of a single sentence—a sentence history or all sentence histories for a text (For the concept of text histories and sentence histories, see Mahlow et al. [11]).

Writing is not only exhausting, but also complicated: the writer types one letter after the other, deletes something, adds something. One action after the other transforms the resulting text. Writing is linear, but only in time. It is non-linear in space. Writing on a keyboard means pressing one key after the other, but moving freely in the *text produced so far* (TPSF). The next letter may appear in a completely different place, the next deleted word may be part of a completely different sentence. This makes analyzing and visualizing writing processes challenging.

When writing on the computer, every keystroke can be recorded. Each action is a data point and contains (1) which key was pressed (2) at what time and (3) at what time it was released, (4) the effect on the text and (5) the position of the cursor in the document. Not all keys leave a direct mark on the text: Shift only capitalizes the next letter, and it must be held.

Switching from typing to deleting or moving the cursor cursor results in a new version, following Mahlow [9]. The actions that led to this version are called *transformation* [14]. These are classified into types: *append*, *insert*, *delete* (at the end of the TPSF), *midlete* (somewhere before the end of the TPSF), *replace*.

Keystroke logging data can be played back, and one can watch oneself writing afterwards. We can count individual elements, characters and words for example, or how often the writer deleted something. We can also tell stories with such data: How did the text come about, what happened to the sentences in between? For these sentence stories and text stories we suggest new types of visualizations.

2 Related Work

The most commonly used tool for keystroke logging in writing research is Inputlog[1] [8], which not only allows researchers to export the logging data in XML format for further processing, but includes several analyses and visualizations. To visualize the writing process as such, the so-called *process graph* as shown in figure 1 is used and widely accepted. However, this visualization needs training to be read and understood. It mixes too many aspects—development of the TPSF, pauses, revisions—without the possibility for a user—be it a researcher or the writer themselves—to exclude some aspects or to adjust parameters. Additionally, it is not very aesthetic from a design viewpoint. Thus, it is rarely used when researchers interact with writers to talk about their writing processes.

There are other proposals, which focus on revisions, such as progression graphs [12], or on time aspects, such as the timeline representation by Wengelin et al. [16]. Caporossi and Leblay [4] developed a graph-based visualization for keystroke logging data recorded with *ScriptLog* [7], which they later developed into

[1] https://www.inputlog.net

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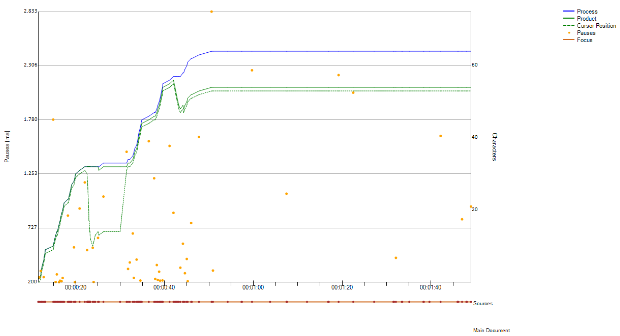

Figure 1: Process graph from a short writing session, produced by Inputlog [own data].

the dedicated graph-based logging application *GenoGraphiX-Log* (GGXLog)[2], but with a focus on the process only. Currently, writing research still lacks appropriate visualizations of the writing process that "represent the problem completely" [2, p. 101]. The "problem" is the fact that writing is both linear and non-linear at the same time. The writer is always free to move around in the TPSF for the next action.

3 Approach

A major design decision, similar to the linguistic analysis proposed by Mahlow [9] and implemented in the *Text History Extraction Tool* (THEtool) [13], was to start from raw data. This allows us to control all processing steps. THEtool accepts keystroke logging data in idfx format, the de-facto standard XML format from Inputlog, which also can be exported from ScriptLog and other logging tools. As we are interested in the development of the text itself, specifically the linguistic units constituting the text, we use information on all sentence histories of a writing session, including detailed information on all transformations involved.

As writers execute actions linearly in time, each version [9] of the growing text is the result of a transformation [14] that takes place at a specific point in the existing text or at the edge of the TPSF. The difference between adjacent versions V_n and V_{n+1} on the surface is the *transforming sequence* [11], which is stored in a hierarchical data structure together with the actions carried out by the writer to achieve V_{n+1} and the classification of the respective transformation according to type and scope [14].

Based on this output data, aggregated by sentence, we developed several new visualization types [10]. Here we present two types.

3.1 Combination of Process and Product

Figure 2 shows an example of visualizing the combination of process and product—the product here is the sequence of individual sentences constituting the text—using the writing session of student E as a beeswarm-like graph. The Y axis lists all sentences in the order in which they appear in the final text—i.e., if sentences are shifted during writing and change positions, they are listed here with the final sentence number. The X axis contains time-related information on transformations: We here only include the *sequence*

[2]https://ggxlog.net/

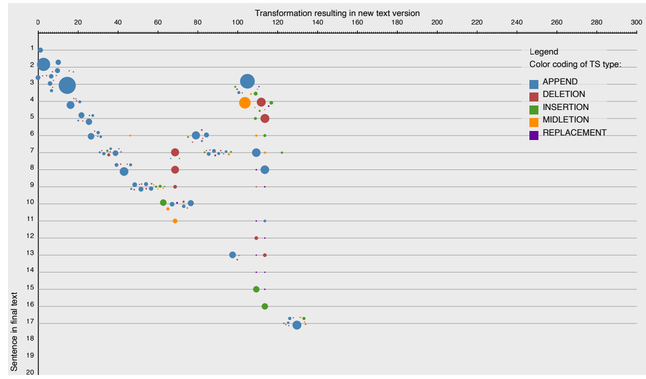

Figure 2: Process graph of sentence histories from the writing session of student E [own data]. Sentences are ordered by appearance in the final text from 1 to 17, starting at the top. The X axis shows transformations in linear order. Each transformation is represented as circle and color-coded by type, the size of each circle encodes the length of this transformation in characters.

of transformations but ignore the duration of a transformation or the time between adjacent transformations, i.e., pauses. This information is available from THEtool, but omitted here.

We use the beeswarm algorithm to arrange the data points indicating a transformation for each sentence in a more readable way to avoid overlap and also show this information for all sentences in the same visualization. Each colored circle represents a transformation resulting in a new text version, the center is placed according to the exact running number of this transformation. The size of the transformation in characters is encoded as the radius of the beeswarm circles.

For linear production—i.e., the writer only edits and revises in the current sentence and never moves to places further away from the current point of inscription—we would see a diagonal line of beeswarm circles. Looking at the sentence lines for writer E, we see that some sentences are visited and changed more than once: Sentence 3 is extended after roughly two thirds of all activities; sentence 7 is created, then revised at four different times while the writer goes back and forth. Following the beeswarm graph, we can follow the writer as they move through the growing text while expanding and shrinking it here and there, and ending with writing the final sentence.

3.2 Sentence Histories

The ultimate goal of our research is the development of a writing model focusing on linguistic structures and units [14, 15]. We focus on sentences, as sentences are an essential linguistic element in communication [3] and thus of texts, a well researched unit in linguistics [1, 5], and a particular research topic in computational linguistics and natural language processing as the development of syntactic parsers like SpaCy [6] shows. Larger structural elements of texts like paragraphs or sections are built from sentences.

For sentence histories, we designed various visualization types that focus on the transformations of each single sentence in the order they appear, but ignore the point in time. Thus, we only visualize all transformations involved for each sentence, but do not indicate whether they occur one after the other, or whether the writer edited other sentence in between. Figure 3 shows this specific visualization. Each sentence has color-coded information on expanding and shrinking transformations. Each expanding transformation (*append* in blue, *insert* in green) shows the length in characters; the length of shrinking transformation (both *deletion, midletion* in red) is color-coded by intensity—the darker the red color in the triangles, the longer the deleted sequence was. We had to find a solution to represent deletions in a static visualization that is not visually misleading with respect to the length of the final sentence. Therefore, the triangle points to the left, the vertical line is placed according to the sequence of transformations.

To further help the user to easily understand the effect of deletions while writing a sentence, the final length of each sentence is the length of the horizontal thin black line. Figure 3 provides more insight into the emergence of the sentences constituting the resulting text. We can immediately spot outliers like sentence 11: this sentence is produced and then never changed again; there is only one single blue bar, and the black line indicating the length of this sentence in the final text has exactly the same length. Sentences with quite a few deletions—e.g., sentence 5—or some longer deletions indicated by darker red-colored deletion triangles—e.g., sentence 24—have rather short result lines. One can easily grasp the difference between the characters produced for a specific sentence compared to the number of characters in the final version of this sentence.

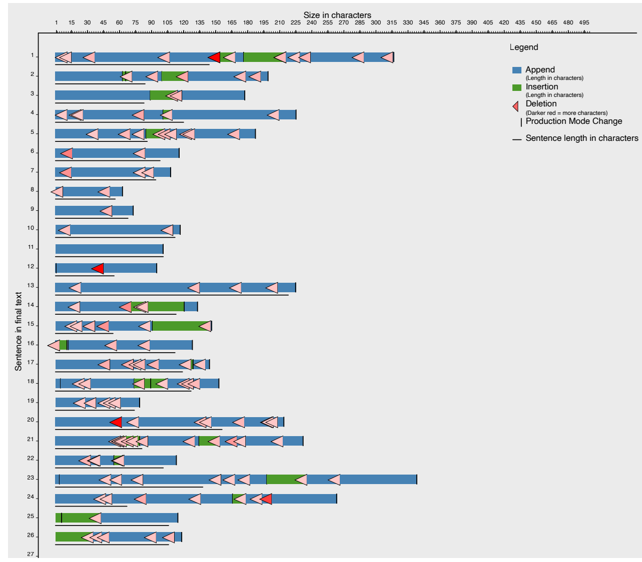

Figure 3: Detailed sentence histories from the writing session of student F [own data]. Sentences are ordered by appearance in the final text from 1 to 26, starting at the top. The X axis shows the size of the displayed units in characters. Each sentence has color-coded information on transformations expanding the sentence (append and insert) and shrinking it (deletion and midletion); transformations are ordered according the sentence history. Each expanding transformation shows the length in characters; the length of shrinking transformation is color-coded. The final length of each sentence is the length of the thin black line.

4 Exploration Study

For the development of the various types of visualizations we used keystroke logging data from writing sessions of nine students (A to F). All students wrote the introduction to their internship reports in just under 40 minutes. Thus, all had the same writing task, but each person produced something different. The students donated their data, logged with ScriptLog, for research purposes.

4.1 Data Preprocessing

The raw keystroke logging data were read in with our analysis tool *THEtool*, aggregated into sentence histories based on transforming sequences, and then output in a simple text format. All visualizations are based exclusively on this output data. All visualizations are produced with our scripts using the JavaScript libraries P5JS[3] and D3[4] as described above.

4.2 Comparison of Multiple Writers

Figure 4 shows the process–product visualizations for three students using the same scale for the X and Y axes. The distinctive cluster of appends and deletions for student B at one thirds of their writing is likely due to fidgeting with the keyboard. Considering the graph produced for writer E in figure 2, it becomes very obvious that the writers spent their time differently and applied different writing strategies to produce the introduction for their internship reports.

[3]https://p5js.org.
[4]https://d3js.org.

Similarly, all visualization types developed can be run for the data of all nine writing sessions. Additionally, all visualization types can be produced for each writing session, allowing to visually find explanations for specific features or for specific difficulties a writer faces and might mention in writing counseling.

4.3 Development of Research Hypothesis

The current visualizations represent a qualitative approach to writing and the writing process of individual writers. They can be used to propose research questions for further exploration on a qualitative—i.e., concerning this particular writer for this particular writing task—or quantitative level—e.g., whether the observed feature is typical for similar writers or a writing genre. The automatic analysis of keystroke logging data does not include information on the quality of the produced texts. From a mechanical and linguistic point of view, some information can be added automatically—e.g., concerning correctnes and completion of single sentences, or concerning coherence and cohesion of sequences of sentences. Adding this information will allow us to distinguish parts of the writing process where writers actually struggled from those where they played around with words and phrases for creative purposes. Both might look identical when only considering keys pressed.

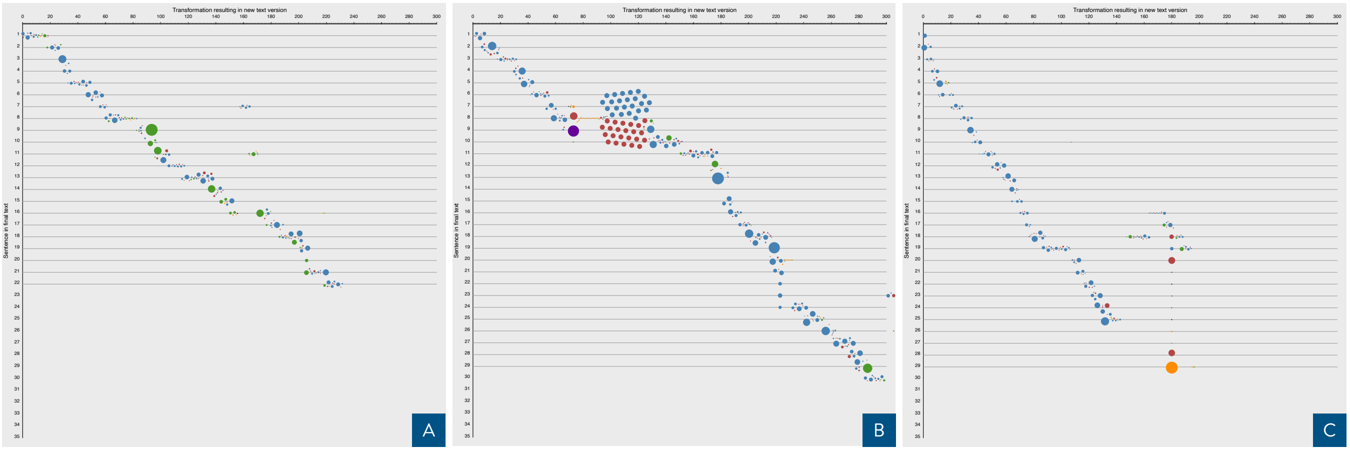

Figure 4: Process graphs of sentence histories from the writing session of three students [own data]. Sentences ordered by appearance in the final text from 1 to *n*, starting at the top. X axis shows transformations in linear order. Each transformation is represented as circle and color-coded by type, the circle size encodes the length of this transformation in characters.

5 Conclusion and Outlook

All visualizations are newly developed and should also find their way into research practice. For the moment, we only have static visualizations to be presented on screen or to be printed out and shown to the writer for further discussion or to the researcher for further investigation. Currently, we are working on the implementation of a digital dashboard that takes the raw keystroke logging data of a writer (or a small group of writers) as input, sends it to THEtool, and then processes the output into the various visualization types. The user will then be able to adjust features and parameters, make exports for presentations and papers, zoom in and out, and display variants of visualization types.

As mentioned in section 3.1, we currently omit time information in the visualization. According to the layer model proposed by Ulasik et al. [14], this information should be added and could be visualized as an additional dimension, preferable manipulable by the user. This would also allow for more interaction with the data and thus hopefully result in more profound understanding of one's own writing habits and strategies.

The next step will be the design of visualization models for exploration with a quantitative focus—e.g., logging data from dozens or hundreds of writers as collected during writing studies in schools for educational studies on the efficacy of writing instructions.

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